

# AKANKSHA SARAN

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## EDUCATION

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| <b>University of Texas at Austin</b><br>PhD in Computer Science                                             | 2015 - 2021 |
| <b>Carnegie Mellon University</b><br>MS in Robotics                                                         | 2013 - 2014 |
| <b>Indian Institute of Technology Jodhpur</b><br>Bachelor of Technology in Computer Science and Engineering | 2009 - 2013 |

## RESEARCH INTERESTS

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Machine Learning, Reinforcement Learning, Human-in-the-loop Learning

## WORK EXPERIENCE

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| <b>Sony AI, Delhi NCR, India</b><br>Research Scientist                                                                                                                                                                                                   | Jan 2026 - Present<br><i>Supervisors: James MacGlashan, Niranjana Pedenekar</i> |
| <ul style="list-style-type: none"><li>Member of the User Engagement Group group working on problems related to interactive machine learning (with a focus on reinforcement learning).</li></ul>                                                          |                                                                                 |
| <b>Sony AI, SF Bay Area, USA</b><br>Research Scientist                                                                                                                                                                                                   | Dec 2023 - Jan 2026<br><i>Supervisors: Peter Wurman, Peter Stone</i>            |
| <ul style="list-style-type: none"><li>Member of the Gaming and Interactive Agents group working on problems related to training generalist AI agents (with a focus on reinforcement learning) for widely popular Sony PlayStation video games.</li></ul> |                                                                                 |
| <b>Microsoft Research, New York City, USA</b><br>Postdoctoral Researcher                                                                                                                                                                                 | Dec 2021 - Dec 2023<br><i>Supervisors: Akshay Krishnamurthy, John Langford</i>  |
| <ul style="list-style-type: none"><li>Member of the Machine Learning and Reinforcement Learning groups working on problems related to sequential decision-making with an eye toward real-world interactive applications.</li></ul>                       |                                                                                 |
| <b>NVIDIA, Seattle, USA</b><br>Graduate Robotics Research Intern                                                                                                                                                                                         | May - Aug 2018<br><i>Supervisors: Jonathan Tremblay</i>                         |
| <ul style="list-style-type: none"><li>Refining human hand and object pose estimation from a monocular camera.</li><li>Real-to-sim robot learning using generative adversarial imitation learning.</li></ul>                                              |                                                                                 |
| <b>Apple, Cupertino, USA</b><br>Graduate Research Intern                                                                                                                                                                                                 | May - Aug 2016<br><i>Supervisors: Shalini Sahoo, Samir Gehani</i>               |
| <ul style="list-style-type: none"><li>Worked with the Interactive Media Group (IMG) to develop algorithms pertaining to user media search.</li></ul>                                                                                                     |                                                                                 |
| <b>Quality of Life Technology Center, CMU, Pittsburgh, USA</b><br>Research Associate Staff                                                                                                                                                               | Jan - June 2015<br><i>Supervisors: Aisling Kelliher, Kris Kitani</i>            |
| <ul style="list-style-type: none"><li>Worked on an inter-disciplinary project for a vision-based stroke rehabilitation system for the home.</li></ul>                                                                                                    |                                                                                 |
| <b>EPFL, Lausanne, Switzerland</b><br>Undergraduate Research Intern                                                                                                                                                                                      | May - July 2012<br><i>Supervisors: Madhura Purnaprajna, Paolo Ienne</i>         |

- Search for heterogeneous logic blocks in FPGA architectures to achieve reduction in area and delay.

## PUBLICATIONS

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$\alpha - \beta$  indicates alphabetical author order, \* indicates equal contribution.

Kim Baraka, Ifrah Idrees, Taylor Kessler Faulkner, Erdem Biyik, Serena Booth, Mohamed Chetounai, Daniel Grollman, Akanksha Saran, Emmanuel Senft, Silvia Tulli, Anna-Lisa Vollmer, Antonio Andriella, Helen Beierling, Tiffany Horter, Jens Kober, Isaac Shiedlower, Matthew Taylor, Sanne Van Waveren, Xuesu Xiao. **Human-Interactive Robot Learning: Definition, Challenges, and Recommendations.** THRI 2025.

Akanksha Saran, Jacob Alber, Cyril Zhang, Ann Paradiso, Danielle Bragg, John Langford. **EyeO: Autocalibrating Gaze Output with Gaze Input for Gaze Typing.** CHI LBW 2025.

Matilda Knierim, Sahil Jain, Murat Han Aydoan, Kenneth D. Mitra, Kush Desai, Akanksha Saran\*, and Kim Baraka\*. **Leveraging Prosody as an Informative Teaching Signal for Agent Learning: Exploratory Studies and Algorithmic Implications.** ICMI 2024 [Oral].

Dipendra Misra\*, Akanksha Saran\*, Tengyang Xie, Alex Lamb, John Langford. **Towards Principled Representation Learning from Videos for Reinforcement Learning.** Short version: Workshop on Reinforcement Learning Beyond Rewards, RLC 2024, Workshop on Open-World Agents, NeurIPS 2024. Full version: ICLR 2024 [Spotlight: 5% acceptance rate].

Akanksha Saran, Safoora Yousefi, Akshay Krishnamurthy, John Langford, Jordan T. Ash. **Streaming Active Learning with Deep Neural Networks.** ICML 2023.

( $\alpha - \beta$ ) Jessica Maghakian, Kishan Panaganti Badrinath, Paul Mineiro, Akanksha Saran, Cheng Tan. **Interaction-Grounded Learning for Recommender Systems.** Short Version: Workshop on Online Recommender Systems and User Modeling, RecSys 2022. Full Version: ICLR 2023.

Harshit Sikchi, Akanksha Saran, Wonjoon Goo, Scott Niekum. **A Ranking Game for Imitation Learning.** Short Version: Deep Reinforcement Learning Workshop, NeurIPS 2022. Full version: TMLR 2023.

Tengyang Xie\*, Akanksha Saran\*, Dylan Foster, Lekan Molu, Ida Momennejad, Nan Jiang, Paul Mineiro, John Langford. **Interaction-Grounded Learning with Action-Inclusive Feedback.** Short Version: Workshop on Complex Feedback for Online Learning, ICML 2022. Full version: NeurIPS 2022.

Akanksha Saran\*, Kush Desai\*, Mai Lee Chang, Rudolf Lioutikov, Andrea Thomaz, Scott Niekum. **Understanding Acoustic Patterns of Human Teachers Demonstrating Manipulation Tasks to Robots.** IROS 2022.

Yuchen Cui, Bo Liu, Akanksha Saran, Stephen Giguere, Peter Stone, Scott Niekum. **Aux-AIRL: End-to-End Self-Supervised Reward Learning for Extrapolating beyond Suboptimal Demonstrations.** Short Version: Workshop on Self-Supervised Learning for Reasoning and Perception, ICML 2021.

Akanksha Saran, Kush Desai, Andrea Thomaz, Scott Niekum. **A Case for Leveraging Human Prosody during Robot Learning.** Short Version: Workshop on Sound in Human-Robot Interaction, HRI 2021.

Akanksha Saran, Ruohan Zhang, Elaine Schaertl Short, Scott Niekum. **Efficiently Guiding Imitation Learning Algorithms with Human Gaze.** Short version: Workshop on Reinforcement Learning in Games, AAAI 2020. Full version: AAMAS 2021.

Ruohan Zhang, Akanksha Saran, Bo Liu, Yifeng Zhu, Sihang Guo, Scott Niekum, Dana H. Ballard and Mary M. Hayhoe. **Human Gaze Assisted Artificial Intelligence: A Review**. IJCAI 2020.

Akanksha Saran, Elaine Schaertl Short, Andrea Thomaz, Scott Niekum. **Understanding Teacher Gaze Patterns for Robot Learning**. Short version: Pioneers Workshop, HRI 2019. Full version: CoRL 2019.

Akanksha Saran, Srinjoy Majumdar, Elaine Schaertl Short, Andrea Thomaz, Scott Niekum. **Human Gaze Following for Human-Robot Interaction**. Short version: Workshop on Social Robots in the Wild, HRI 2018. Full version: IROS 2018.

Akanksha Saran, Branka Lakic, Srinjoy Majumdar, Juergen Hess, Scott Niekum. **Viewpoint Selection for Visual Failure Detection**. IROS 2017.

Akanksha Saran, Damien Teney, Kris Kitani. **Hand Parsing for Fine-Grained Recognition of Human Grasps in Monocular Images**. Short version: Workshop on Observing and Understanding Hands in Action, CVPR 2015. Full version: IROS 2015.

## WORKING PAPERS

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Red-Teaming Vision-Language-Action Models via Quality Diversity Prompt Generation for Robust Robot Policies. Expected submission in 2026.

Symmetry Discovery for Reinforcement Learning. Expected submission in 2026.

Stabilizing RLHF for LLMs. Expected submission in 2026.

Coachable agents for interactive gameplay. Expected submission in 2026.

## PATENTS

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Akanksha Saran, Jacob Alber, Danielle Bragg, Cyril Zhang, John Langford. **EyeO: Autocalibrating Gaze Output with Gaze Input**. Dec 2025

## TEACHING EXPERIENCE

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**University of Texas at Austin** Fall 2018  
*Course: Artificial Intelligence*

**University of Texas at Austin** Fall 2015  
*Course: Elements of Computers and Programming*

**Carnegie Mellon University** Fall 2014  
*Course: Computational Photography*

## INVITED TALKS

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HRI Student Speaker Series, Colorado School of Mines August 2020

Workshop on Solutions for Socially Intelligent HRI in Real World Scenarios March 2021

Microsoft Research NYC May 2021

IBM Research May 2021

Facebook Reality Labs June 2021

Google Research June 2021

University of Southern California April 2023

Sony AI Research April 2023

Yale University April 2023

Microsoft Research Redmond May 2023

University of Washington May 2023

Toyota Research Institute  
OpenAI  
Adobe Research

June 2023  
August 2023  
December 2025

## HONORS AND AWARDS

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|----------------------------------------------------------------------------|------------|
| EECS Rising Star                                                           | 2023       |
| AAMAS Scholar                                                              | 2021       |
| Google's CS Research Mentorship Program (CSRMP) Scholar                    | 2021       |
| Top 10 Finalist for the 3MT Competition, UT Austin                         | 2020       |
| Women in Engineering Leadership Scholarship, UT Austin                     | 2020       |
| HRI Pioneer                                                                | 2019       |
| Student and Developing Countries Travel Award, IROS                        | 2018       |
| ISSS Financial Aid Scholarship, UT Austin                                  | 2018, 2020 |
| Adobe Research Fellowship Finalist                                         | 2018       |
| CRA-W Graduate Women Cohort                                                | 2017       |
| NSF Doctoral Consortium Award, IROS                                        | 2017       |
| 3rd Place, RoboCup@Home Domestic Standard Platform League                  | 2017       |
| UT Professional Development Award for travel to HRI, RSS, IROS, CoRL       | 2017-2020  |
| Young Women Researcher Grant, RSS                                          | 2014       |
| GSA/Provost Research and Conference Funding Award, CMU                     | 2014       |
| EPFL Research Scholarship (Declined)                                       | 2013       |
| Google Anita Borg Memorial Scholarship Asia Pacific                        | 2013       |
| TOEFL Scholarship                                                          | 2013       |
| American Alumni Association Scholarship                                    | 2013       |
| President's Gold Medal, IIT Jodhpur                                        | 2013       |
| Proficiency Medal, IIT Jodhpur                                             | 2013       |
| Overall Performance Award, IIT Jodhpur                                     | 2011-2013  |
| USC Viterbi Research Scholarship (Declined)                                | 2012       |
| IIT-EPFL Internship Scholarship (Among 7 students selected all over India) | 2012       |

## REVIEWING

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HRI 2019, RSS 2019, ICRA 2019, HRI 2020, RSS 2020, NeurIPS 2020 (top 10%), AAAI 2021, ICLR 2021, IROS 2021, CoRL 2021, HRI 2022, IROS 2022, CoRL 2022, JMLR 2023, HRI 2024, ICML 2024 workshops, CoRL 2024, NeurIPS 2024, NeurIPS 2024 workshops, NeurIPS 2025 workshops, ICRA 2026, RSS 2026

## MENTEES AT UT AUSTIN AND MICROSOFT RESEARCH

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|                                                                        |                        |
|------------------------------------------------------------------------|------------------------|
| Jessica Maghakian (MSR intern, now at Goldman Sachs)                   | August 2022 - May 2023 |
| Harshit Sikchi (UT Austin PhD student, now at OpenAI)                  | Sept 2021 - Feb 2023   |
| Kishan Panaganti Badrinath (MSR intern, now at Tencent AI Labs)        | June 2022 - Dec 2022   |
| Sahil Jain (UT Austin undergrad, now at Sony AI)                       | Aug 2020 - May 2022    |
| Kush Desai (UT Austin undergrad, now at xAI)                           | Sept 2018 - Feb 2021   |
| Srinjoy Majumdar (UT Austin undergrad, now CEO Shasta Health (YC S23)) | Jan 2017 - Feb 2018    |

## LEADERSHIP

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|------------------------------------------------------------------|------|
| <b>Area Chair</b>                                                | 2026 |
| <i>International Conference on Machine Learning (ICML), 2026</i> |      |

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| <b>Area Chair (Top 10%)</b><br><i>Annual Conference on Neural Information Processing Systems (NeurIPS), 2025</i>        | 2025        |
| <b>Area Chair</b><br><i>International Conference on Machine Learning (ICML), 2025</i>                                   | 2025        |
| <b>Workshop Co-Organizer</b><br><i>Reinforcement Learning beyond Rewards, RLC 2024</i>                                  | 2024        |
| <b>Workshop Co-Organizer</b><br><i>Interactive Learning with Implicit Human Feedback, ICML 2023</i>                     | 2023        |
| <b>Workshop Co-Organizer</b><br><i>All Things Attention: Bridging Different Perspectives on Attention, NeurIPS 2022</i> | 2022        |
| <b>Research Discussion Moderator</b><br><i>Workshop on Human-Interactive Robot Learning, HRI 2022</i>                   | 2022        |
| <b>Workshop Co-Organizer</b><br><i>Program Chair, HRI Pioneers 2021</i>                                                 | 2020 - 2021 |
| <b>Workshop Co-Organizer</b><br><i>Advances and Challenges in Imitation Learning for Robotics, RSS 2020</i>             | 2020        |
| <b>Coordinator</b><br><i>Girl Day, UT Austin</i>                                                                        | 2020        |
| <b>Workshop Co-Organizer</b><br><i>Web and Publicity Chair, HRI Pioneers 2020</i>                                       | 2019 - 2020 |
| <b>Executive Committee Member</b><br><i>Graduate Representative Association of Computer Science (GRACS), UT Austin</i>  | 2017 - 2018 |
| <b>Secretary</b><br><i>Graduate Women in Computing (GWC), UT Austin</i>                                                 | 2016 - 2017 |
| <b>Coordinator</b><br><i>Entrepreneurship Meet, IIT Jodhpur</i>                                                         | 2012, 2013  |
| <b>Assistant Coordinator</b><br><i>Literature Club, IIT Jodhpur</i>                                                     | 2012, 2013  |
| <b>Festival Co-Organizer</b><br><i>Techno-Cultural Festival, IIT Jodhpur</i>                                            | 2012, 2013  |

## OUTREACH

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|---------------------------------------------------------------------------|------|
| <b>Panelist</b><br><i>IIT Jodhpur Undergraduate Orientation</i>           | 2021 |
| <b>Panelist</b><br><i>UT Women in Computer Science Research Hackathon</i> | 2021 |
| <b>Volunteer</b><br><i>AAAI Undergraduate Consortium</i>                  | 2021 |
| <b>Volunteer</b><br><i>UTCS Robotics Summer Camp</i>                      | 2019 |
| <b>Volunteer</b><br><i>UT Explore</i>                                     | 2018 |

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| <b>Volunteer and Mentor</b><br><i>Women in Engineering Program (WEP), UT Austin</i> | 2017-2020  |
| <b>Volunteer</b><br><i>UTCS Hour of Code</i>                                        | 2016, 2019 |
| <b>Mentor</b><br><i>#include Fellowship Program</i>                                 | 2014, 2016 |
| <b>Volunteer</b><br><i>Women@SCS, CMU</i>                                           | 2014       |
| <b>Mentor</b><br><i>Programming Club, IIT Jodhpur</i>                               | 2011-2012  |
| <b>Volunteer</b><br><i>Weekly Rural Medical Health Program, IIT Kanpur</i>          | 2009-2010  |